

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I V.Thanusri (192472265) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Thanusri

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.**
- (b) Identify key challenges in detecting boundaries.**
- (c) Apply a suitable edge detection method.**
- (d) Justify your selection with logical reasoning.**
- (e) Present your explanation clearly and systematically.**

ANS:-Object Boundary Detection in Medical Images

(a) Interpretation of the Scenario and Importance of Boundary Detection

In medical imaging, different tissues and organs often appear close together and may have similar intensity values. Boundary detection is used to identify the edges between these tissues accurately. It helps doctors and medical systems distinguish one structure from another.

Importance:

- Helps in identifying tumors, organs, and abnormal tissues.
- Improves the accuracy of medical diagnosis.
- Assists in image segmentation for treatment planning.
- Enables precise measurement of tissue size and shape.
- Supports computer-aided diagnosis systems

(b) Key Challenges in Detecting Boundaries

Several factors make boundary detection difficult in medical images:

- Presence of noise in medical scans.
- Low contrast between neighboring tissues.
- Blurred or weak edges.
- Variations in image quality and resolution.
- Complex shapes and overlapping structures.
- Uneven illumination and artifacts.

(c) Suitable Edge Detection Method

A suitable method for this scenario is the **Canny Edge Detection Algorithm**.

Steps involved:

1. Apply Gaussian filtering to remove noise.
2. Compute intensity gradients to detect possible edges.
3. Perform non-maximum suppression to thin the edges.
4. Use double thresholding to identify strong and weak edges.
5. Apply edge tracking by hysteresis to produce the final boundaries.

(d) Justification for Selecting Canny Edge Detection

The Canny edge detector is preferred because:

- It provides accurate and well-defined boundaries.
- It effectively removes image noise before edge detection.
- It detects weak and strong edges reliably.
- It produces thin and continuous edges.
- It reduces false edge detection compared to simpler methods.
- It is widely used in medical image processing applications.

Since medical images require high precision, Canny edge detection offers a good balance between sensitivity and accuracy.

(e) Systematic Presentation

Aspect	Explanation
Scenario	Detect boundaries between tissues in medical images.
Importance	Helps diagnosis, segmentation, and treatment planning.
Challenges	Noise, low contrast, blurred edges, and complex structures.
Method Used	Canny Edge Detection Algorithm.
Reason for Selection	Accurate, noise-resistant, and produces clear boundaries.
Outcome	Improved identification of tissue boundaries and better medical image analysis.

Conclusion

Boundary detection plays an important role in medical image analysis by clearly separating tissues and anatomical structures. Among various techniques, the Canny edge detection method is highly suitable because it provides accurate, continuous, and noise-resistant boundary detection, making it effective for medical imaging applications.

2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.**
- (b) Identify key features enabling separation.**
- (c) Apply a suitable segmentation technique.**
- (d) Justify your decision logically.**
- (e) Present your response clearly.**

Simple Segmentation

(a) Interpretation of the Scenario and Segmentation Requirements

The image contains objects that are clearly different from the background based on their intensity (brightness). The goal of segmentation is to separate the objects from the background accurately.

Segmentation Requirements:

- Clearly distinguish objects from the background.
- Group pixels with similar intensity values.
- Produce accurate object boundaries.
- Minimize errors during object extraction.

(b) Key Features Enabling Separation

The following features help in separating the objects:

- Difference in pixel intensity between objects and background.
- Uniform brightness levels within the objects.
- High contrast between foreground and background.
- Minimal overlap of intensity values.

(c) Suitable Segmentation Technique

A suitable technique is **Threshold-Based Segmentation (Thresholding)**.

Procedure:

1. Analyze the image intensity values.
2. Select an appropriate threshold value.
3. Classify pixels:
 - Pixels above the threshold are treated as objects.
 - Pixels below the threshold are treated as background.
4. Generate the segmented image.

(d) Justification for the Selection

Threshold-based segmentation is chosen because:

- It is simple and easy to implement.
- It works effectively when objects and background have distinct intensity levels.
- It requires less computational effort.
- It provides accurate separation for high-contrast images.
- It is one of the most commonly used techniques for basic image segmentation problems.

Since the objects are clearly distinguishable from the background by intensity, thresholding is the most appropriate and efficient method.

(e) Clear and Systematic Presentation

Aspect	Explanation
Scenario	Objects are distinguishable from the background by intensity.
Requirement	Separate objects accurately from the background.
Key Features	Intensity difference and high contrast.
Technique Used	Threshold-Based Segmentation.
Reason for Selection	Simple, fast, and effective for intensity-based images.
Outcome	Accurate extraction of objects from the image.

Conclusion

Threshold-based segmentation is highly suitable for images where objects are clearly distinguishable by intensity. It offers a simple, efficient, and accurate approach for separating foreground objects from the background.

3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- (a) Interpret the scenario and explain segmentation challenges.**
- (b) Identify key issues such as intensity overlap.**
- (c) Apply an appropriate advanced segmentation method.**
- (d) Justify how the method handles complexity.**
- (e) Provide a structured and clear explanation.**

a) Interpretation of the Scenario and Segmentation Challenges

In this scenario, the image contains multiple regions whose intensity values overlap. This makes it difficult to separate objects using simple methods because different regions may appear similar in brightness.

Segmentation Challenges:

- Multiple objects have similar intensity values.
- Boundaries between regions are unclear.
- Objects may overlap or touch each other.
- Noise and variations in texture can affect segmentation accuracy.

(b) Key Issues Such as Intensity Overlap

The major issues include:

- Overlapping intensity values among different regions.
- Low contrast between neighboring objects.
- Complex shapes and sizes of objects.
- Presence of image noise and artifacts.
- Difficulty in accurately identifying object boundaries.

(c) Appropriate Advanced Segmentation Method

A suitable advanced segmentation technique is **Watershed Segmentation**.

Procedure:

1. Preprocess the image to reduce noise.
2. Compute the gradient image to highlight object boundaries.
3. Identify markers for foreground and background regions.
4. Apply the watershed algorithm to separate different regions.
5. Obtain the segmented image with clearly defined boundaries.

(d) Justification for the Selected Method

Watershed segmentation is appropriate because:

- It effectively separates touching and overlapping objects.
- It uses gradient information to detect boundaries accurately.
- It handles complex images with multiple regions.
- Marker-based watershed reduces over-segmentation.
- It provides precise segmentation even when intensity values overlap.

Unlike simple thresholding methods, watershed segmentation considers both region information and object boundaries, making it suitable for complex segmentation tasks

(e) Structured and Clear Presentation

Aspect	Explanation
Scenario	Image contains multiple regions with overlapping intensity values.
Challenges	Intensity overlap, unclear boundaries, and noise.
Key Issues	Similar brightness levels and complex object structures.
Technique Used	Watershed Segmentation.
Reason for Selection	Handles overlapping regions and accurately detects boundaries.
Outcome	Improved segmentation of complex images.

Conclusion

Watershed segmentation is an effective advanced technique for complex images where intensity values overlap. It accurately separates multiple regions and provides reliable segmentation results, making it well suited for challenging image-processing applications.

4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.**
- (b) Identify key issues affecting boundary continuity.**
- (c) Apply a suitable method to improve edge connectivity.**
- (d) Justify your approach logically.**
- (e) Present your explanation clearly.**

ANS:

a) Interpretation of the Scenario and Why Edges Are Broken

After edge detection, the boundaries of objects appear incomplete or disconnected. This means that the edge detection algorithm failed to identify all parts of the object boundary continuously.

Reasons for Broken Edges:

- Image noise affects edge detection.
- Weak or low-contrast boundaries are not fully detected.
- Improper threshold values are used.
- Variations in illumination or texture cause interruptions.

(b) Key Issues Affecting Boundary Continuity

The major issues include:

- Noise present in the image.
- Weak edge responses.
- Gaps between detected edge pixels.
- Inaccurate threshold selection.
- Irregular object boundaries.

(c) Suitable Method to Improve Edge Connectivity

A suitable method is **Morphological Closing**.

Procedure:

1. Perform edge detection on the image.
2. Apply morphological closing (dilation followed by erosion).
3. Fill small gaps between disconnected edge segments.
4. Obtain continuous and smoother object boundaries.

(d) Justification for the Selected Method

Morphological closing is chosen because:

- It effectively connects broken or disconnected edges.
- It fills small gaps without significantly altering the object shape.
- It improves boundary continuity and smoothness.
- It is computationally simple and widely used in image processing.

By joining nearby edge pixels, morphological closing produces more complete and reliable object boundaries.

(e) Clear and Structured Presentation

Aspect	Explanation
Scenario	Object boundaries are discontinuous after edge detection.
Problems	Noise, weak edges, and gaps in detected boundaries.
Technique Used	Morphological Closing.
Reason for Selection	Connects broken edges and improves boundary continuity.
Outcome	Continuous and smoother object boundaries.

Conclusion

Morphological closing is an effective technique for improving edge connectivity in images. It fills small gaps and produces continuous object boundaries, making edge detection results more accurate and useful for further image analysis.

5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpret the scenario and define the combined requirement.

(b) Identify key issues involving noise and edge preservation.

(c) Apply an appropriate sequence of techniques.

(d) Justify the order and selection with strong reasoning.

(e) Present your response in a clear and structured format.

(a) Interpretation of the Scenario and Combined Requirement

In an industrial inspection system, images may contain noise that can affect defect detection. At the same time, important object edges must be preserved because defects are often identified from their boundaries and shapes.

Combined Requirements:

- Remove image noise effectively.
- Preserve important edges and fine details.
- Improve the accuracy of defect detection.
- Ensure reliable image analysis for inspection purposes.

(b) Key Issues Involving Noise and Edge Preservation

The major issues are:

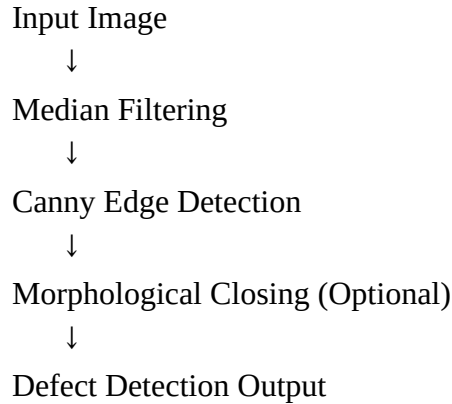
- Noise may create false defects.
- Excessive smoothing can blur important edges.
- Weak defect boundaries may become difficult to detect.
- Accurate edge preservation is essential for identifying defects.

(c) Appropriate Sequence of Techniques

A suitable sequence of image processing techniques is:

1. **Median Filtering** – removes noise while preserving edges.
2. **Canny Edge Detection** – accurately detects object and defect boundaries.
3. **Morphological Operations (Closing)** – improves edge continuity if necessary.

Processing Flow:



(d) Justification for the Order and Selection

- **Median Filtering** is applied first to eliminate noise without significantly affecting edge information.
- **Canny Edge Detection** is applied after noise removal because it provides accurate and continuous edge detection.
- **Morphological Closing** is used at the end to connect small gaps in detected edges and improve boundary quality.

This sequence is logical because edge detection performs better on noise-free images, and post-processing improves the final edge representation for defect analysis.

Advantages of the selected approach:

- Reduces false detections caused by noise.
- Preserves critical defect boundaries.
- Produces accurate and continuous edges.
- Improves the overall reliability of industrial inspection systems.

(e) Clear and Structured Presentation

Aspect	Explanation
Scenario	Industrial inspection requires noise removal and edge preservation.
Requirements	Remove noise while maintaining important object boundaries.
Issues	Noise, blurred edges, and inaccurate defect detection.

Techniques Used Median Filtering → Canny Edge Detection → Morphological Closing.

Reason for Selection Provides effective noise removal and accurate edge preservation.

Outcome Reliable and accurate defect detection.

Conclusion

For industrial defect inspection, applying Median Filtering followed by Canny Edge Detection and Morphological Closing provides an effective solution. This approach removes noise, preserves important edges, and improves the accuracy and reliability of defect detection.